Lab4Visual

July 25, 2025

Exploring 1D Data Visualizations

[43]:

```
# Import required libraries at the top
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load your dataset
df = pd.read_csv("train_and_test2.csv")

# Display first few rows
df.head()
```

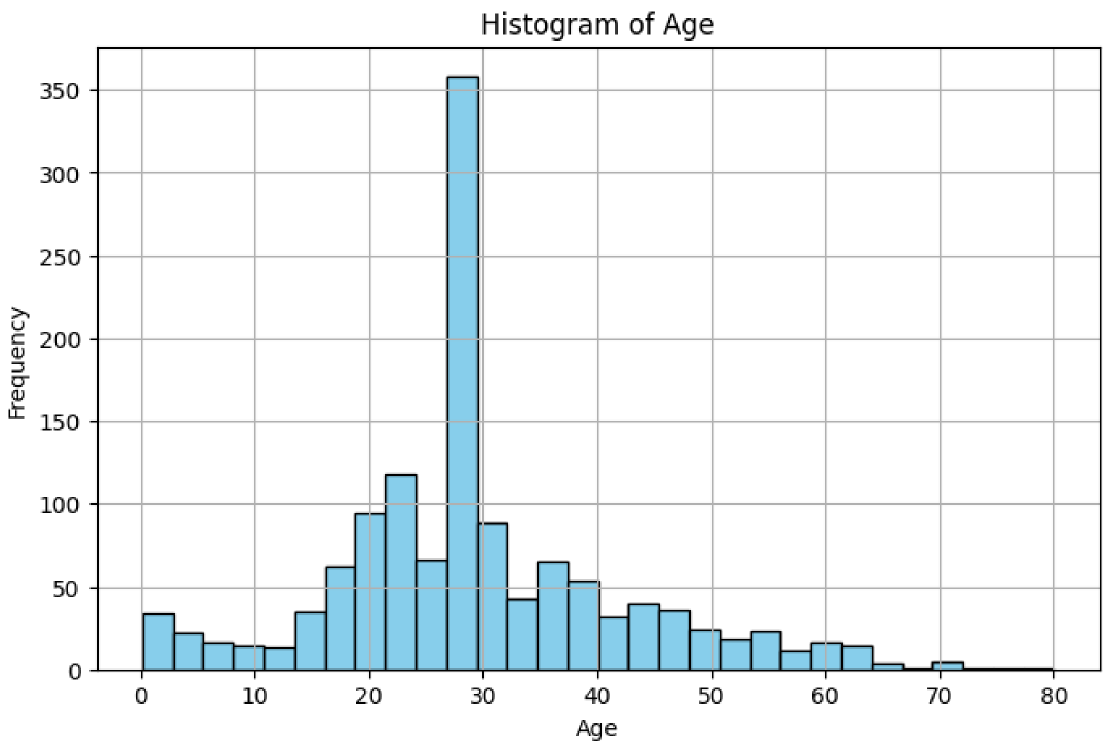
```
Index(['Passengerid', 'Age', 'Fare', 'Sex', 'sibsp', 'zero', 'zero.1',
       'zero.2', 'zero.3', 'zero.4', 'zero.5', 'zero.6', 'Parch', 'zero.7',
       'zero.8', 'zero.9', 'zero.10', 'zero.11', 'zero.12', 'zero.13',
       'zero.14', 'Pclass', 'zero.15', 'zero.16', 'Embarked', 'zero.17',
       'zero.18', '2urvived'],
      dtype='object')
```

1. Histograms (for Numerical Data)

Standard Histogram – Age

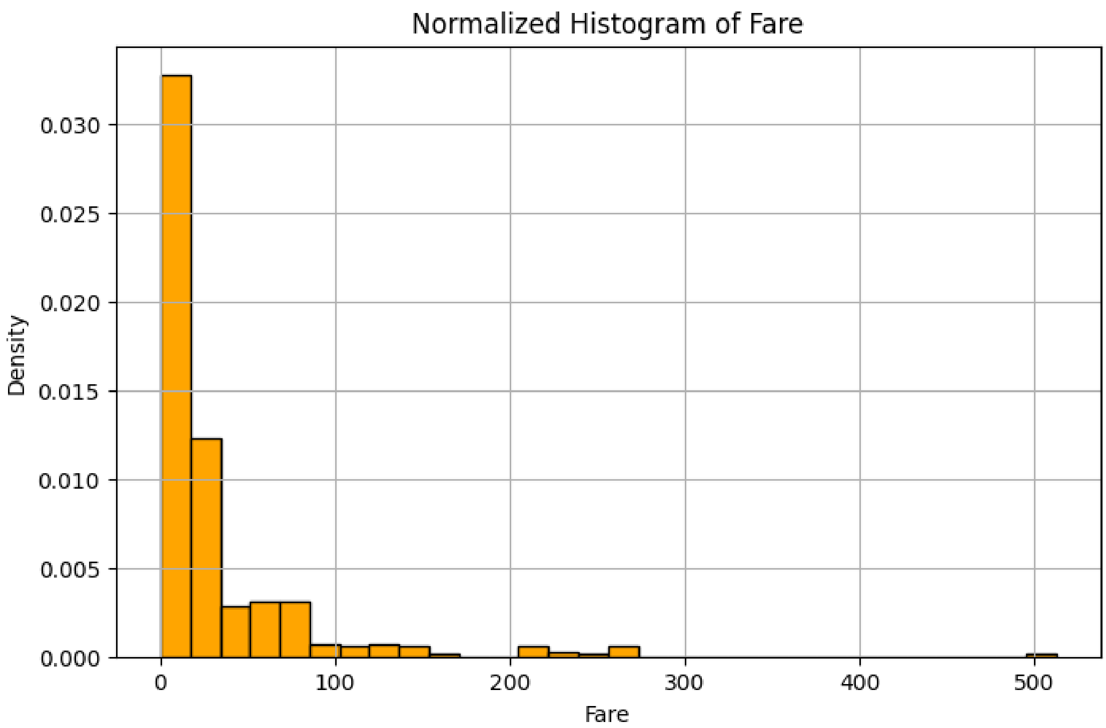
[32]:

```
plt.figure(figsize=(8, 5))
plt.hist(df['Age'].dropna(), bins=30, color='skyblue', edgecolor='black')
plt.title('Histogram of Age')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
```



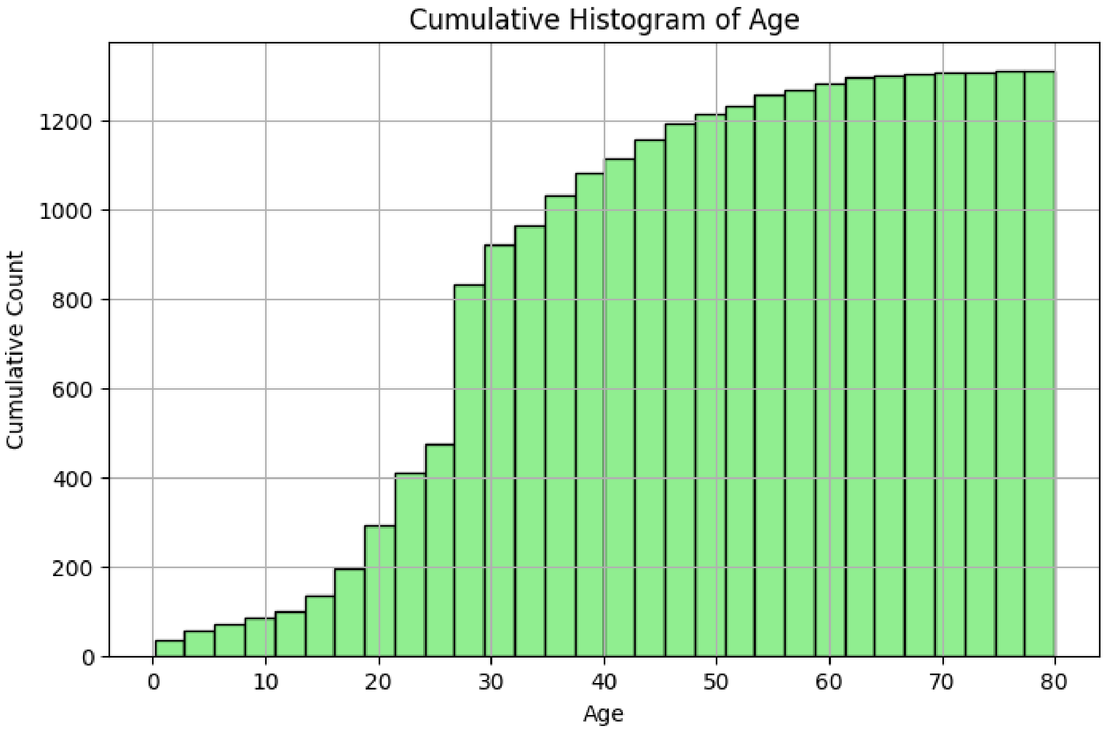
1.2 Normalized Histogram – Fare

[33]:
```
plt.figure(figsize=(8, 5))
plt.hist(df['Fare'], bins=30, density=True, color='orange', edgecolor='black')
plt.title('Normalized Histogram of Fare')
plt.xlabel('Fare')
plt.ylabel('Density')
plt.grid(True)
plt.show()
```



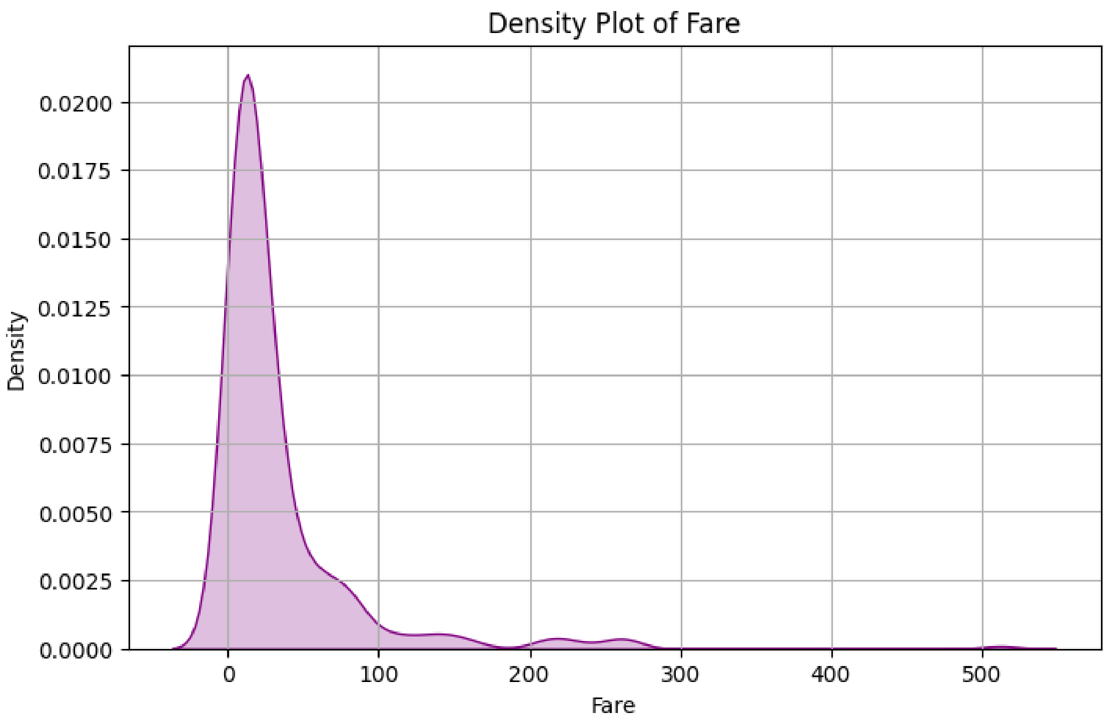
1.3 Cumulative Histogram – Age

[34]:
```
plt.figure(figsize=(8, 5))
plt.hist(df['Age'].dropna(), bins=30, cumulative=True, color='lightgreen',␣
  ↪edgecolor='black')
plt.title('Cumulative Histogram of Age')
plt.xlabel('Age')
plt.ylabel('Cumulative Count')
plt.grid(True)
plt.show()
```



1.4 KDE / Density Plot (Seaborn)

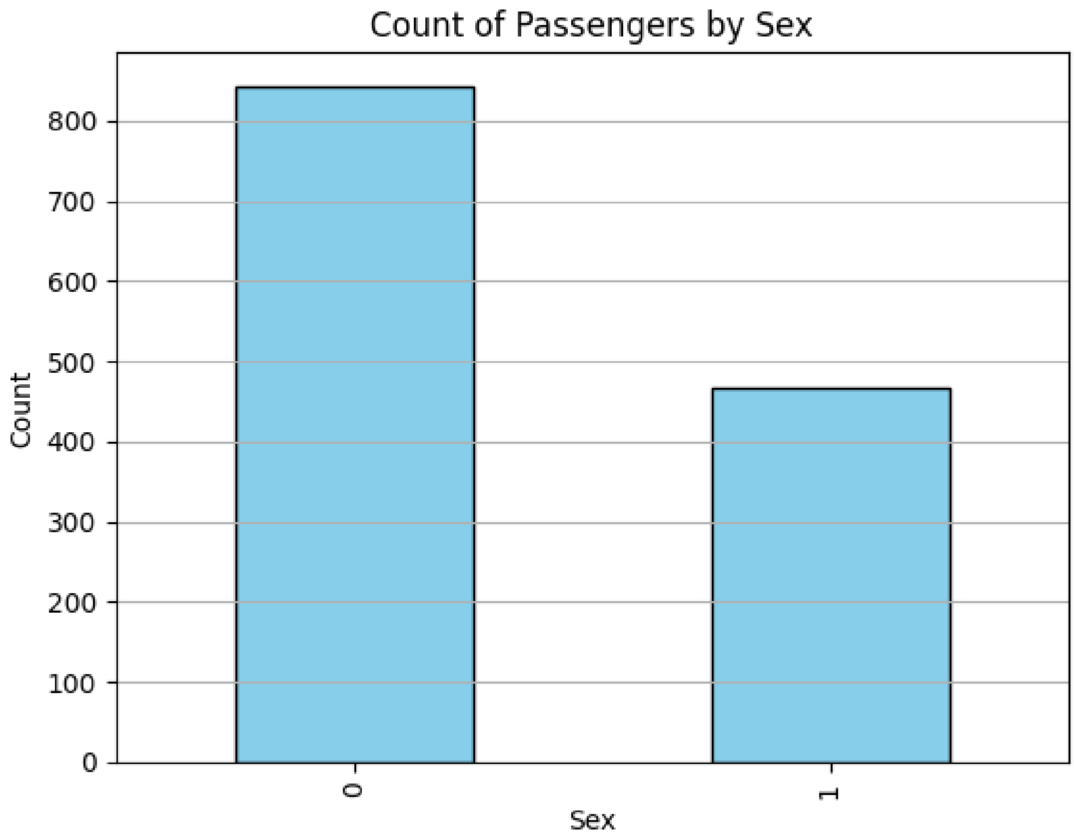
[35]:
```
plt.figure(figsize=(8, 5))
sns.kdeplot(df['Fare'], fill=True, color='purple')
plt.title('Density Plot of Fare')
plt.xlabel('Fare')
plt.ylabel('Density')
plt.grid(True)
plt.show()
```



2. Bar Plots (for Categorical Data)

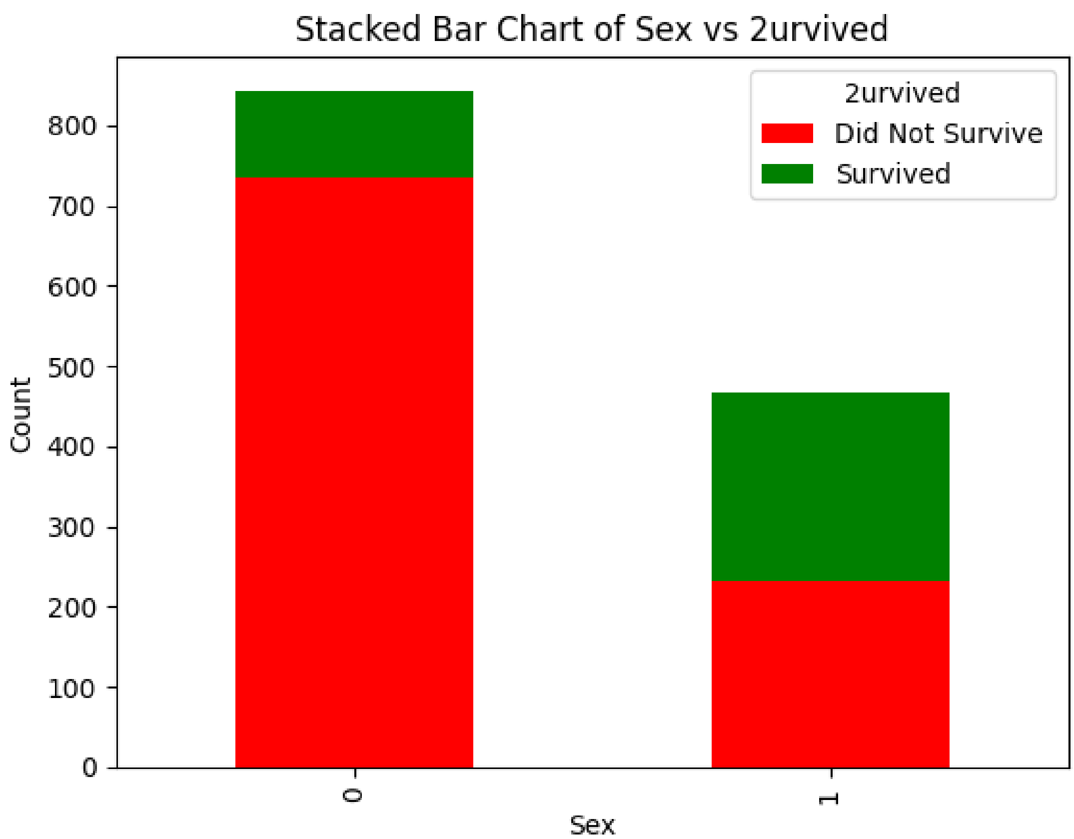
2.1 Vertical Bar Chart – Sex

[36]:
```
df['Sex'].value_counts().plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Count of Passengers by Sex')
plt.xlabel('Sex')
plt.ylabel('Count')
plt.grid(True, axis='y')
plt.show()
```



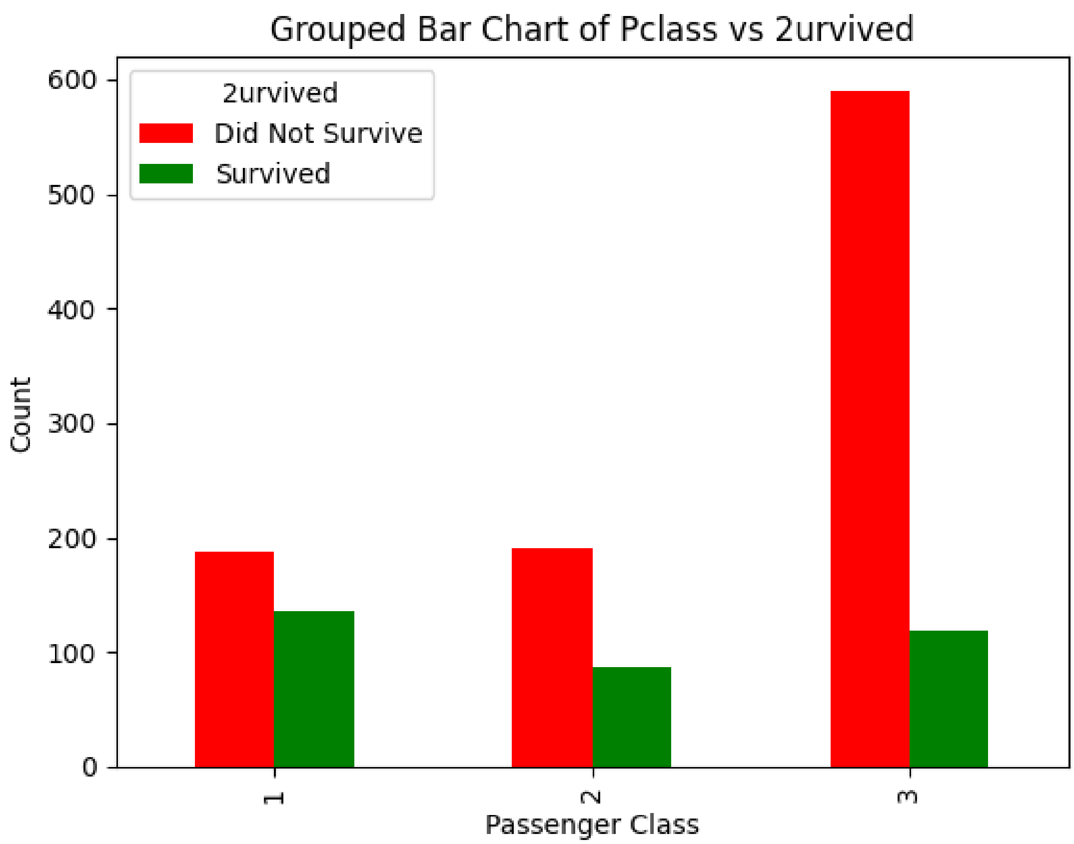
2.2 Stacked Bar Chart – Sex vs. Survival

[45]:
```
pd.crosstab(df['Sex'], df['2urvived']).plot(kind='bar', stacked=True,␣
↪color=['red', 'green'])

plt.title('Stacked Bar Chart of Sex vs 2urvived')
plt.xlabel('Sex')
plt.ylabel('Count')
plt.legend(title='2urvived', labels=['Did Not Survive', 'Survived'])  # Adjust␣
↪labels if needed
plt.show()
```



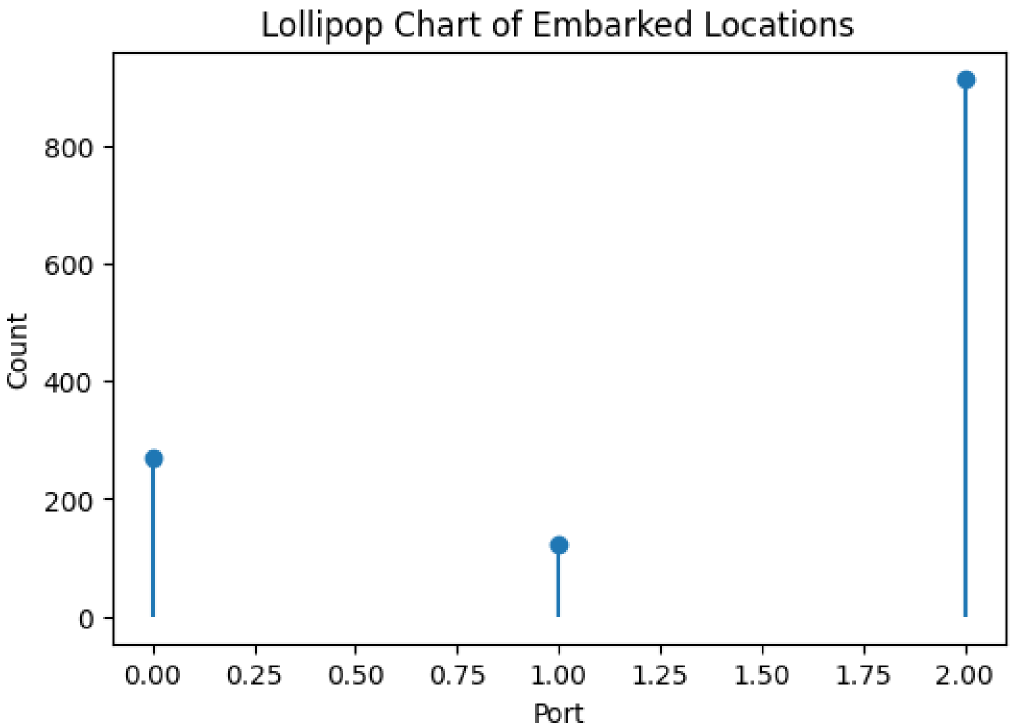
2.3 Grouped Bar Chart – Pclass vs. Survived

[47]:
```
pd.crosstab(df['Pclass'], df['2urvived']).plot(kind='bar', stacked=False,␣
↪color=['red', 'green'])

plt.title('Grouped Bar Chart of Pclass vs 2urvived')
plt.xlabel('Passenger Class')
plt.ylabel('Count')
plt.legend(title='2urvived', labels=['Did Not Survive', 'Survived'])  # Adjust␣
↪labels as needed
plt.show()
```



2.4 Lollipop Chart – Embarked

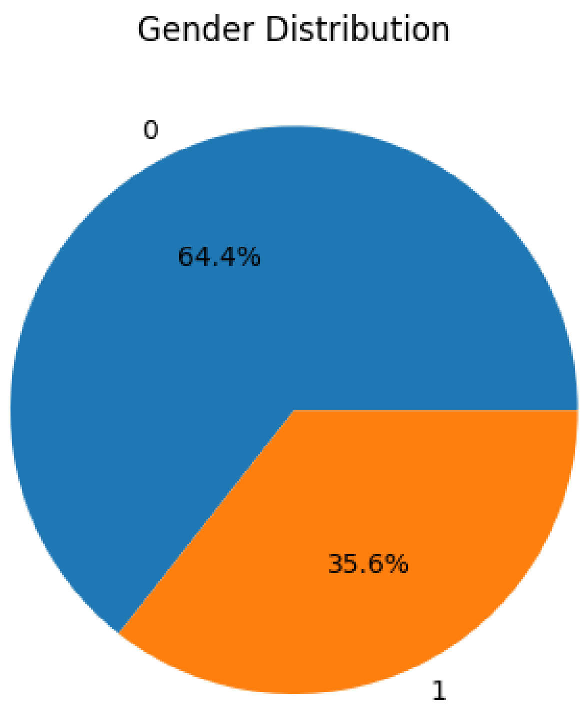
[48]:
```
counts = df['Embarked'].value_counts()
plt.figure(figsize=(6,4))
plt.stem(counts.index, counts.values, basefmt=" ")
plt.title("Lollipop Chart of Embarked Locations")
plt.xlabel("Port")
plt.ylabel("Count")
plt.show()
```



3. Pie Charts (for Category Proportion)

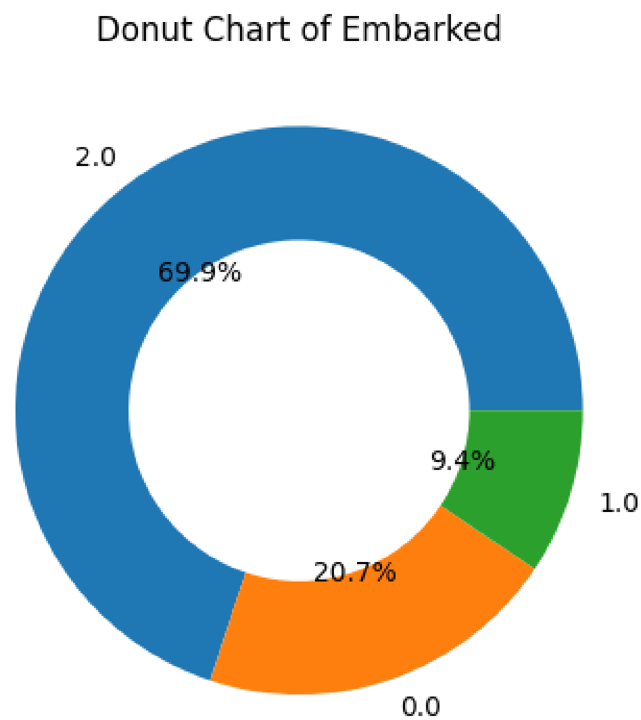
3.1 Standard Pie Chart – Sex

[49]:
```
df['Sex'].value_counts().plot.pie(autopct='%1.1f%%')
plt.title("Gender Distribution")
plt.ylabel('')
plt.show()
```



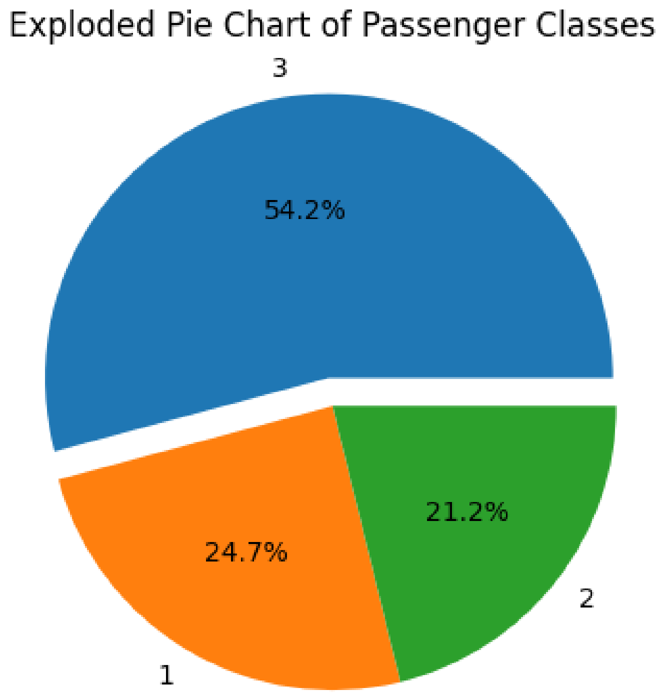
3.2 Donut Chart – Embarked

[50]:
```
embarked_counts = df['Embarked'].value_counts()
plt.pie(embarked_counts, labels=embarked_counts.index, autopct='%1.1f%%',␣
↪wedgeprops=dict(width=0.4))
plt.title("Donut Chart of Embarked")
plt.show()
```



3.3 Exploded Pie Chart – Pclass

[51]:
```
pclass_counts = df['Pclass'].value_counts()
explode = (0.1, 0, 0)
plt.pie(pclass_counts, labels=pclass_counts.index, explode=explode, autopct='%1.
↪1f%%')
plt.title("Exploded Pie Chart of Passenger Classes")
plt.show()
```



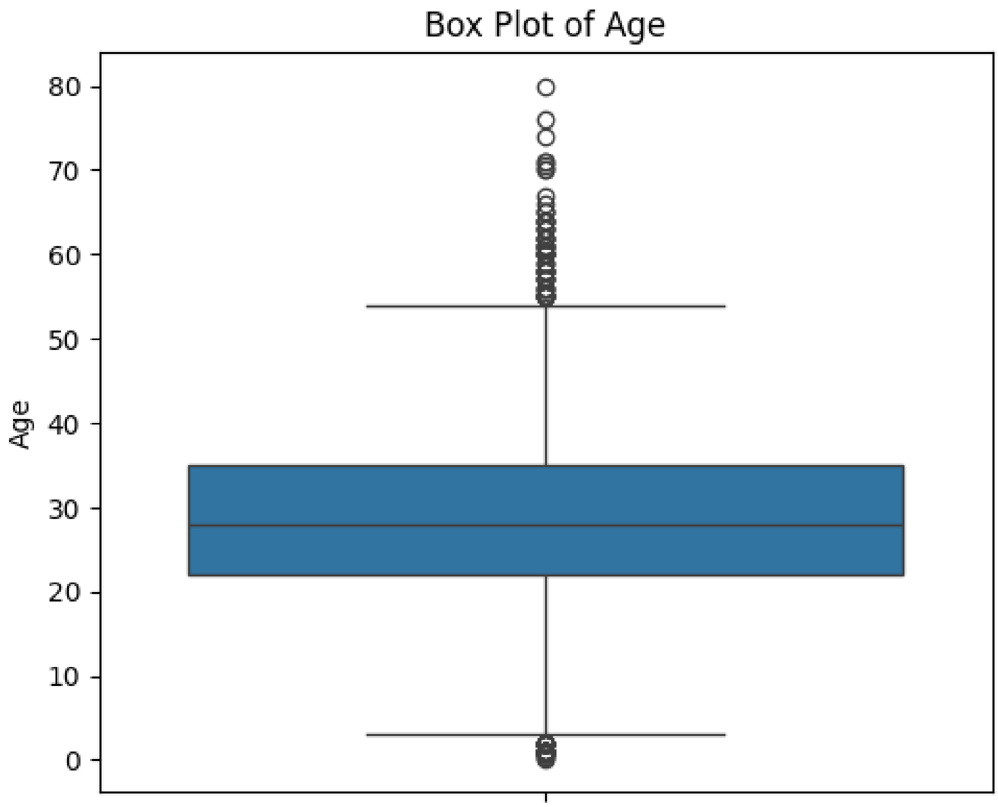
3.4 Nested Pie (Sunburst style) – Sex and Survived

[53]:
```
import plotly.express as px

fig = px.sunburst(df, path=['Sex', '2urvived'], values=None)
fig.update_layout(title="Nested Pie (Sunburst) of Sex and 2urvived")
fig.show()
```

4. Box Plots (for Statistical Summary)

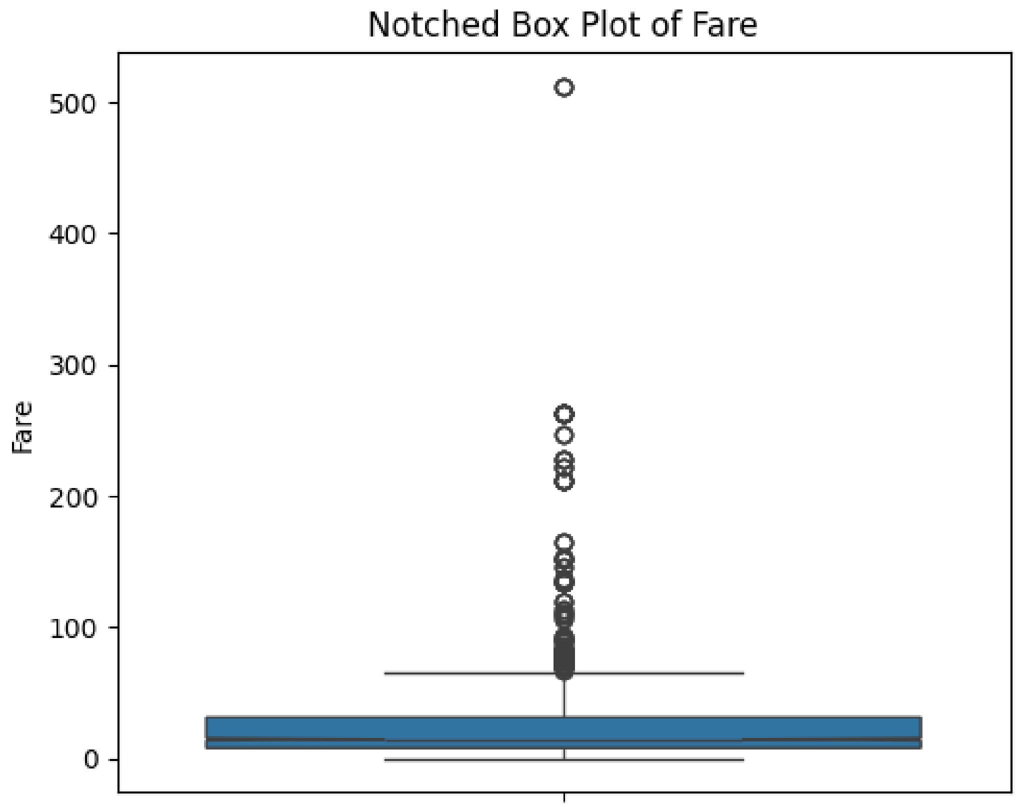
4.1 Standard Box Plot – Age

[54]:
```
plt.figure(figsize=(6,5))
sns.boxplot(y='Age', data=df)
plt.title("Box Plot of Age")
plt.show()
```



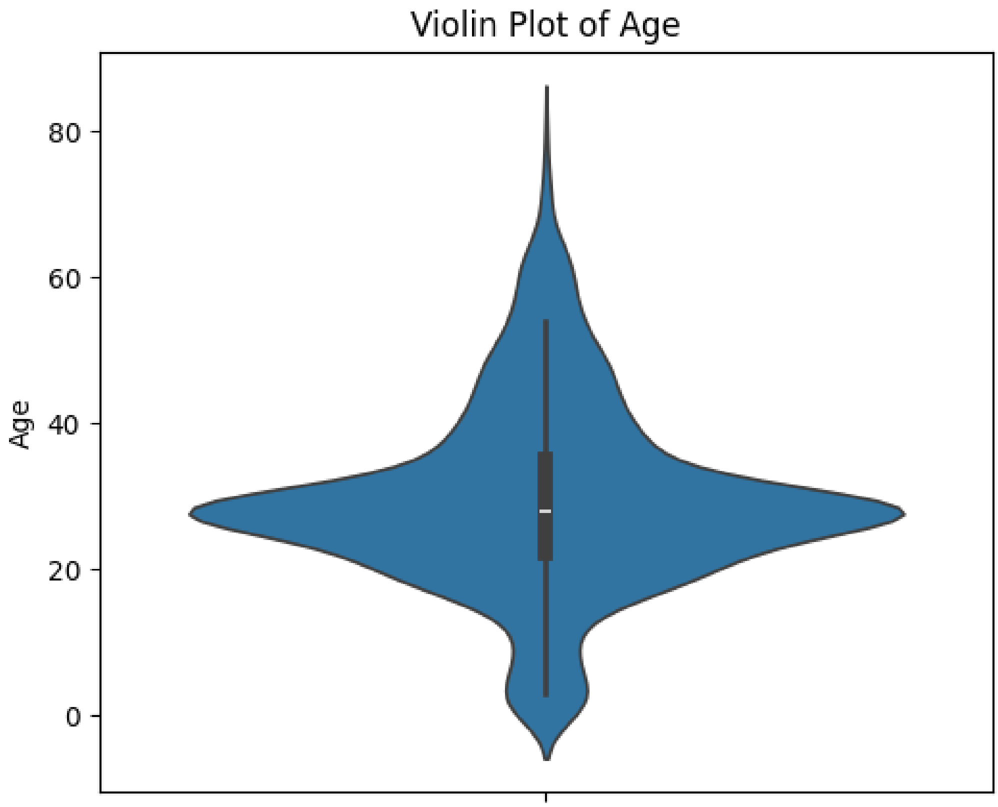
4.2 Notched Box Plot – Fare

[55]:
```
plt.figure(figsize=(6,5))
sns.boxplot(y='Fare', data=df, notch=True)
plt.title("Notched Box Plot of Fare")
plt.show()
```



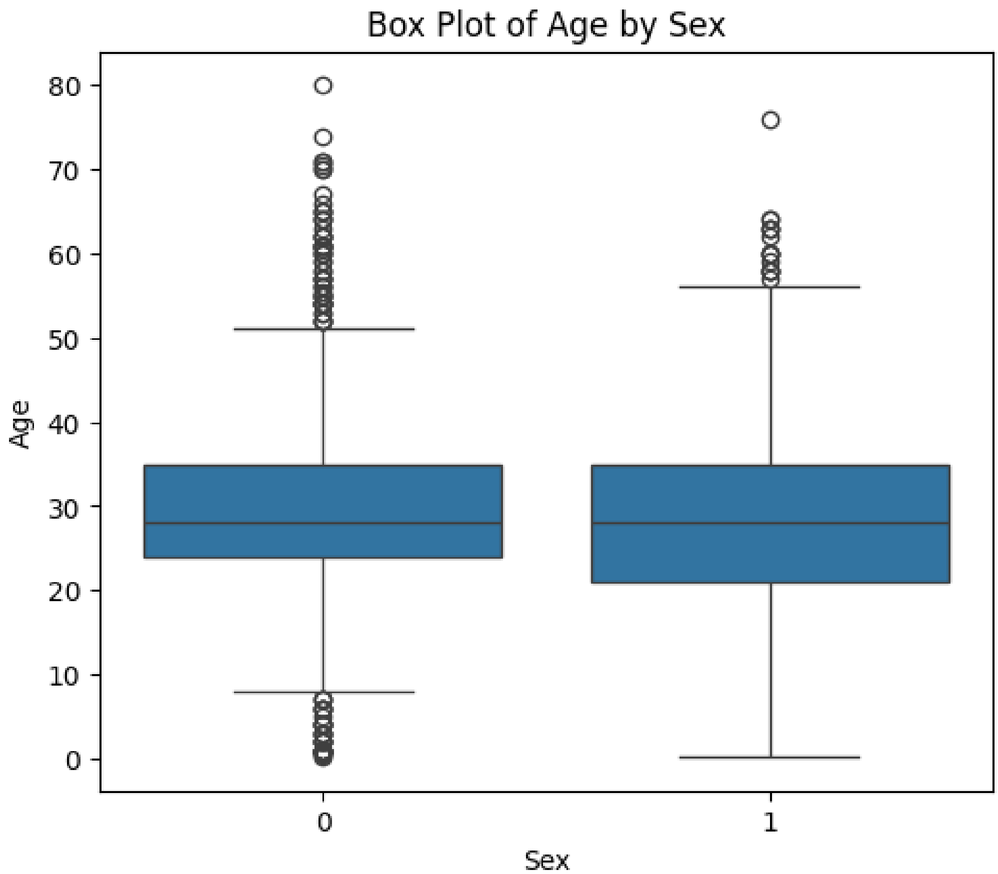
4.3 Violin Plot – Age

[56]:
```
plt.figure(figsize=(6,5))
sns.violinplot(y='Age', data=df)
plt.title("Violin Plot of Age")
plt.show()
```



4.4 Side-by-side Box Plot – Age by Sex

[57]:
```
plt.figure(figsize=(6,5))
sns.boxplot(x='Sex', y='Age', data=df)
plt.title("Box Plot of Age by Sex")
plt.show()
```


Box Plot of Age by Sex
80
70
60
50
40
30
20
10
0
Age
0
1
Sex